

Independent Computational Verification

Paper II — Theorem 1.1: the global maximum over chordal graphs

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Artifact: F_theorem_1_1_global/audit_F_global.py **Verdict:** PASS

Claim

$\max_{|V(G)|=n, G \text{ chordal}} \Phi_\tau(G) = \left\lfloor \frac{(2n+1)^2}{24} \right\rfloor$, attained by a complete-split graph.

F1 — EXHAUSTIVE (n ≤ 7)

All non-isomorphic chordal graphs on 1..7 vertices (NetworkX atlas), exact Φ_τ by LP.

n	$\max \Phi_\tau$	floor	argmax is complete-split
1	0.0	0	yes
2	1.0	1	yes
3	2.0	2	yes
4	3.0	3	yes
5	5.0	5	yes
6	7.0	7	yes
7	9.0	9	yes

Every $n \leq 7$: the maximum **equals** the floor, and a maximizer **is** complete-split. PASS.

F2 — SAMPLED (n = 8..12)

For each n : **3000** random chordal graphs (built so their neighborhood-at-insertion is a clique $\Rightarrow$ chordal by construction; RNG seed 20260709) **plus** the full complete-split family. Attainment of the floor is certified **exactly** via the Section-6 closed form on the complete-split family.

n	floor	max sampled Φ_τ	samples exceeding floor	complete-split attains floor
8	12	9.0	0	yes
9	15	10.666667	0	yes
10	18	11.0	0	yes
11	22	13.333333	0	yes
12	26	15.0	0	yes

No sampled chordal graph exceeded the floor in any case; the floor is attained exactly by a complete-split. (Random samples fall strictly below the floor, as expected — the extremizer is a specific $S_{p,q}$ unlikely to be hit at random.) PASS.

Scope statement (honest)

Section F is **exhaustive for** $n \leq 7$ and a **random sample for** $8 \leq n \leq 12$ (clearly labelled as such); the $n \geq 8$ part is strong corroborating evidence, not an exhaustive proof over all chordal graphs on n vertices. Runtime 50.66 s.

Verdict

PASS. The global-maximum identity is confirmed exhaustively up to 7 vertices and by large random sampling with exact attainment thereafter.

*This document reports an **independent computational verification** of the stated claims: each result was recomputed from scratch by a self-contained script (exact linear programming via SciPy/HiGHS for the fractional triangle-cover number, exact rational arithmetic for the integer optimization, and exhaustive enumeration of non-isomorphic graphs via the NetworkX atlas where indicated). This is computational evidence corroborating the analytic proof; it is not itself a formal proof. The machine-checked formal proof is the Lean development, which is conditional on the classical chordal-structure theory.*